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CFD-based investigation of sparger position and aeration rate in inclined flat plate photobioreactors

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ABSTRACT

Optimizing photobioreactor (PBR) design is essential for improving the productivity and energy efficiency of microalgal cultivation systems. This study employed Computational Fluid Dynamics (CFD) simulations to assess the hydrodynamic performance of an inclined flat plate PBR for *Arthrospira platensis* cultivation. The CFD model, validated against experimental data (maximum discrepancy: 8.4 %, $R^2 = 0.81$), reliably predicted biomass productivity and internal flow dynamics. Five sparger configurations and four aeration rates were investigated for their effects on radial velocity, turbulence kinetic energy (TKE), and dead zone formation.

The results highlighted the hydrodynamic advantages of the rear-most sparger position (R). At 0.21 vvm (volume of air per volume of culture per minute), position R achieved a radial velocity of 0.125 $m \cdot s^{-1}$, a TKE of 4.32×10^{-3} $m^2 \cdot s^{-2}$, and a dead zone fraction of 18.23 %, closely matching the middle sparger position at 0.23 vvm. Notably, 0.23 vvm represented the highest tolerable aeration rate experimentally, as exceeding this threshold induced shear-related mechanical stress, negatively impacting microalgal cell integrity and reducing productivity. Thus, sparger position R provided equivalent mixing at reduced aeration, lowering energy demand and operational stress.

The inclined geometry enhanced flow uniformity and turbulence, particularly with rearward sparger placement. Integrating dead zone analysis with velocity and TKE metrics, this study offers a validated framework for optimizing sparger design in inclined flat plate PBRs. These findings have significant implications for improving energy efficiency and scalability in laboratory and industrial scale microalgal cultivation systems.

1. Introduction

Microalgae have emerged as promising sources of bioactive compounds, with *Arthrospira platensis* (*A. platensis*) being one of the most extensively studied species due to its high phycocyanin content and diverse biological properties [1,2]. Commonly known as *Spirulina*, *A. platensis* is widely recognized as a "superfood" due to its exceptional nutritional profile, which includes proteins, vitamins, essential fatty acids, amino acids, minerals, and phytonutrients [3–5]. Traditionally cultivated in open ponds [6], this species faces limitations in such systems, including low biomass productivity, contamination risks, and significant water loss [7,8]. As a result, closed photobioreactors (PBRs) have gained attention for their ability to enhance biomass yield while mitigating these challenges [9].

Among various PBR designs, flat plate PBRs are particularly attractive due to their high surface-to-volume ratio (SVR), scalability, and ease of maintenance [10,11]. These features make flat plate systems ideal for microalgae cultivation, especially for strains like *A. platensis* that thrive under controlled conditions [12].However, despite advancements since the 1950s [13], challenges such as scalability, operating costs, and optimization of PBR designs remain unresolved [14]. These techno-economic limitations are especially critical in the context of industrial-scale algal biofuel production [15].

Effective cultivation requires addressing critical parameters such as light intensity, nutrient availability and distribution, mixing, and hydrodynamic performance, all of which significantly influence microalgal growth and productivity [16–18]. Outdoor systems face added complexity due to light fluctuations and thermal variability, which significantly affect productivity [19]. Additionally, limited mixing in traditional PBRs often leads to dead zones and light attenuation, reducing overall efficiency [20,21]. Improving internal hydrodynamics has thus become a central focus in reactor design [22].

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Inclined flat plate PBRs (IFP-PBRs) have emerged as a promising configuration for improving light distribution and mass transfer [23,24]. However, variations in inclination angle and flow dynamics can significantly affect mixing patterns, light-dark cycles, and nutrient distribution, thus influencing biomass yield and photosynthetic performance [25,26]. Although field-scale experiments are useful, they often suffer from limited reproducibility, high costs, and uncontrollable environmental conditions [27–29]. As such, there is a growing need for modelling tools that can integrate hydrodynamics, light, mass transfer, and growth kinetics.

Recent reviews have highlighted that realistic simulation models must account for key biological and environmental factors, including light inhibition, temperature response, and mixing regimes, which were often oversimplified in earlier models [30,31]. While advanced kinetic and CFD models are increasingly capable, they remain underutilized in large-scale applications due to high computational costs and limited validation.

Computational Fluid Dynamics (CFD) is a powerful tool for simulating microalgal cultivation systems, optimizing operational parameters, and aiding photobioreactor design and scale-up with cost-effective and efficient workflows [32–34]. Advanced implementations of CFD have coupled fluid mechanics with dynamic photosynthesis models to evaluate spatial productivity patterns in raceways [35]. Other studies have shown that CFD can optimize reactor mixing and gas exchange [36,37], with growing emphasis on sparger design, which is a critical component that influences turbulence, bubble size, and flow distribution [37,38].

Sparger design and placement significantly impact the internal hydrodynamics of PBRs, affecting not only mixing but also energy consumption and cellular stress [39,40]. Adjusting parameters like hole size, orientation, and sparger position can reduce dead zones and shear while maintaining uniform flow patterns [38,41]. However, current research in this area has largely focused on vertical PBRs [38,42,43], with limited attention to inclined flat-plate systems. Moreover, sparger optimization in these works typically lacked experimental validation and rarely considered metrics such as dead zone quantification [39,44].

Several studies have examined the importance of sparger design in optimizing the hydrodynamics of photobioreactors. For instance, Kulkarni et al. [45] demonstrated that sparger design parameters such as hole size, number, and gas flow rate significantly impact the hydrodynamics and light transmission in flat-plate PBRs, resulting in improved mixing and enhanced biomass productivity. Ranganathan et al. [33] further emphasized the importance of sparger configurations in minimizing dead zones, which can lead to inefficient nutrient distribution and suboptimal biomass yield. These findings underscore the need for optimized sparger design to achieve better mixing efficiency and reduce energy consumption in PBRs.

Previous research on sparger optimization has primarily focused on vertical flat-plate systems, where several studies, including those by Ali et al. [38] and Kulkarni et al. [45], explored how gas flow rate and hole diameter affect flow distribution and mixing efficiency. However, these studies have largely overlooked inclined flat-plate systems. This study extends these findings by investigating the impact of sparger placement in inclined flat-plate configurations, revealing that the inclination angle, combined with sparger placement, plays a critical role in flow uniformity and mixing efficiency. The gravity-assisted rise of bubbles in inclined systems significantly influences mixing and nutrient distribution, providing new insights into optimizing such systems.

Building on these advances, this study investigates the impact of air injection flow rates and sparger positioning on the hydrodynamic performance and biomass productivity of a flat plate PBR. The research integrates experimental findings on *A. platensis* cultivation with advanced CFD simulations, which are validated against experimental data ($R^2 = 0.81$). By exploring the role of sparger placement in optimizing mixing efficiency, reducing dead zones, and enhancing productivity, this work aims to advance the design and operational sustainability of flat plate PBRs.

In contrast to prior studies, this study introduces several novel contributions to the literature. First, it is among the first to optimize sparger positioning specifically in inclined flat-plate photobioreactors, a configuration rarely studied in CFD works. Second, the study demonstrates that rearward sparger placement can achieve equivalent mixing at a reduced aeration rate (0.21 vvm), which minimizes energy use while avoiding shear-induced cellular stress, which is offering a balance between performance and sustainability. Third, unlike many purely numerical studies, the CFD model is experimentally validated ($R^2 = 0.81$), enhancing the reliability of the results. Additionally, the study goes beyond traditional parameters by incorporating radial velocity, turbulence kinetic energy (TKE), and dead zone quantification, providing a more comprehensive view of flow behaviour. Finally, the study accounts for realistic outdoor operating conditions, considering regional constraints (e.g., harsh Australian climates) and applying the findings to *A. platensis*, making them directly relevant to scale-up and industrial applications.

By bridging the gap between sparger design in vertical and inclined PBRs, this research offers valuable insights for optimizing the performance of large-scale microalgae cultivation systems. The results provide a foundation for future studies that could incorporate baffles or advanced flow control elements to further enhance mixing performance and energy efficiency.

2. Materials and methods

2.1. Experimental methods and data collection

To validate the Computational Fluid Dynamics (CFD) model, experimental data were gathered from the cultivation of *Arthrospira platensis* (*A. platensis* MUR 126) at Murdoch University's Algae Research and Development Innovation Hub. The strain was initially maintained in 25 L open paddle wheel-driven microponds using sterilized Zarrouk medium [46]. It was then transferred to 140 L flat plate photobioreactors (PBRs) with dimensions of 1.26 m × 1.25 m (L x H) and an illuminated surface area of 1.58 m^2. The culture depth, or light-path, was maintained at 0.10 m for optimal light penetration. The PBRs featured Infrared Reflective Films (IRF) on their illumination surfaces to enhance temperature regulation and optimize light interception for *A. platensis* cultivation [26,47].

During the two-week experimental phase, microalgae suspension was consistently mixed using filter-sterilized ambient air introduced through 1.20 m long ceramic diffusers positioned at the base of each PBR. The air, pumped using a PondOne O_2 Plus 8000 air pump, had flow rates precisely regulated at aeration rates from 0.17 vvm to 0.23 vvm. Lower aeration rates (<0.17 vvm) were associated with insufficient mixing and reduced biomass productivity, while exceeding 0.23 vvm induced significant physiological stress on *A. platensis*, as reflected by declines in effective quantum yield (*f'q/f'm*) [26]. The initial culture concentration was set at 0.6 g/L, strictly adhering to the protocols of Nwoba et al. [47]. Continuous monitoring of culture temperature was conducted using an underwater temperature logger, and Photosynthetic Active Radiation (PAR) light readings were taken using a PHOTOBIO Advanced Quantum PAR Meter. Biomass productivity and specific growth rate were determined following the procedures outlined by Herold et al. [48].

The primary objective was to collect data on *A. platensis* growth and productivity under various mixing rates. Biomass productivity at different mixing rates, provided a robust dataset for CFD model validation. In addition, the physical dimensions, configuration, and boundary conditions of this experimental rig were directly used to construct the CFD geometry and simulation scenarios. This methodology established a baseline for simulation and facilitated the exploration of different sparger configurations to optimize mixing and enhance microalgal productivity in flat plate PBRs.

2.2. CFD model development

The CFD model was developed using ANSYS Fluent 2022 R1 to simulate the fluid dynamics and hydrodynamic behaviour within the flat plate photobioreactors (PBRs). The reactor geometry was precisely recreated in ANSYS DesignModeler, reflecting the actual experimental dimensions (1.25 m height, 1.26 m width, and 0.10 m depth). Key features such as air sparger positioning and injection points were included to evaluate the impact of air injection on fluid mixing and turbulence. The computational domain was discretised using a structured mesh, providing uniform grid spacing and ensuring accurate resolution of flow gradients and turbulent structures. Special attention was given to modelling the air-liquid interface to capture multiphase interactions. Additionally, a User-Defined Function (UDF) was integrated to incorporate biomass growth kinetics influenced by irradiance and biomass concentration, allowing coupling between hydrodynamics and algal productivity.

2.2.1. Mesh generation and grid independence

Mesh generation is a fundamental aspect of Computational Fluid Dynamics (CFD) simulations, ensuring accurate resolution of flow dynamics and numerical stability. To evaluate grid independence, three meshes with sizes of 2,802,009, 5,471,316, and 8,729,043 cells were tested using the middle sparger position (C) at an aeration rate of 0.21 vvm. The simulated *A. platensis* productivity was compared against experimental data, and the percentage discrepancy was calculated as the relative error between the two. The resulting maximum relative discrepancies were 7.25 %, 4.92 %, and 4.50 %, respectively. The mesh with 5,471,316 cells was selected for subsequent simulations as it achieved an optimal balance between accuracy and computational cost. While the finer mesh (8,729,043 cells) marginally reduced the discrepancy to 4.50 %, its computational time was approximately 1.8 times higher, making it less practical for extensive simulations.

Convergence was achieved when the scaled residuals for all governing equations dropped below 10^{-5} and the productivity values stabilized over consecutive iterations, ensuring reliable and accurate results. A summary of mesh sizes, productivity discrepancies, and computational times is presented in Table 1. The selected mesh resolution ensures a reliable and efficient framework for simulating hydrodynamics and biomass productivity in inclined flat plate photobioreactors.

Simulations were performed on the Nebula Supercomputer at the Pawsey Supercomputing Research Centre, equipped with dual Intel® Xeon® Gold 6248R CPUs, featuring a total of 96 logical processors. This high-performance computing setup facilitated the resolution of large-scale meshes and ensured accurate and efficient simulations of hydrodynamic performance in photobioreactors.

2.2.2. Boundary and solver setup

The simulations were performed using the pressure-based solver in ANSYS Fluent to resolve the fluid dynamics and multiphase interactions within the flat plate photobioreactor (PBR). A transient approach was adopted to capture the dynamic behaviour of the air-liquid flow, with a time step size of 0.001 s to ensure temporal accuracy. A velocity inlet condition was applied at the sparger, with air injection flow rates matching those used in the experimental setup. At the top of the reactor, a pressure outlet condition was implemented, allowing only the gas phase (dispersed phase) to escape, while the liquid phase (continuous phase) was constrained within the reactor. All reactor walls were treated as no-slip boundaries to account for the adherence between the fluid and the reactor surfaces.

The Eulerian–Eulerian multiphase model was selected to represent air–liquid interactions due to its computational efficiency and demonstrated applicability in previous photobioreactor (PBR) studies [49,50]. A constant bubble diameter of 3 mm was specified in the simulations, corresponding to the average bubble size produced by the ceramic diffuser used in the experimental setup, consistent with values reported in similar CFD studies [51–53]. The Eulerian–Eulerian approach is commonly used when gas volume fractions are relatively high and phase interactions dominate system behaviour, and its suitability has been further highlighted in our recent review of CFD applications in PBR engineering [34].

For turbulence closure, the standard k–ε model with standard wall functions was employed. This model provides a balance between numerical stability and accuracy in fully turbulent, aerated multiphase flows, particularly in confined domains like flat panel PBRs [54]. While more advanced models such as Reynolds Stress Models (RSM) or Large Eddy Simulations (LES) could offer higher fidelity, they impose significant computational burdens that are not practical for extensive simulation series [55]. As the simulated productivity results deviated by less than 7 % on average from experimental values, the selected turbulence model was deemed sufficiently accurate. The governing hydrodynamic and turbulence model equations are not repeated here but can be found in Cho et al. [56], a comprehensive CFD study specifically focused on bioreactor mixing dynamics.

The operating Reynolds numbers corresponding to the simulated aeration conditions were in the range of 6800–13,400 across the studied aeration rates. These values are well above the transitional regime and confirm fully turbulent conditions, thereby validating the use of the k–ε turbulence closure. This range is also consistent with values reported in previous CFD–PBR studies [56–58].

Simplifications included assuming constant fluid properties, neglecting bubble coalescence and breakup, and treating the system as isothermal, which are standard practices in CFD simulations of air–sparged photobioreactors [57,59,60]. Overall, the model assumptions and boundary configurations were closely aligned with the experimental system. Their validity is supported not only by literature but also by the strong match between simulation and experimental productivity data, as discussed in Section 3.1.

2.2.3. Microalgal growth modelling

The growth of microalgae in the photobioreactor was modelled using a User-Defined Function (UDF) developed in ANSYS Fluent. This UDF calculates the biomass growth rate as a function of irradiance, biomass concentration, and reactor geometry, based on a modified light-limited photosynthesis model. The selected model was adapted from literature, simplifying the growth kinetics to focus on light as the primary limiting factor for photosynthesis [26,61]. The growth rate was derived from the oxygen production rate, which is stoichiometrically linked to biomass formation.

The UDF incorporates a two-part calculation: first, determining the local irradiance distribution within the reactor using the Lambert-Beer law, and second, calculating the photosynthetic rate at each point based on the available light intensity. The direct and diffuse components of irradiance were considered, with the following equation used for light intensity at a given depth:

$$G(z) = G_D(z) + G_d(z) \tag{1}$$

where $G_D(z)$ and $G_d(z)$ represent the direct and diffuse light intensities, respectively, and are functions of the biomass concentration and reactor depth. The direct ($G_D(z)$) and diffuse ($G_d(z)$) light intensities were computed based on absorption and scattering properties of the culture,

Table 1
Comparison of mesh sizes, discrepancy percentages, computational times, and the selected configuration.

Mesh size (cells)	Discrepancy (%)	Computational time (hours)	Selected
2,802,009	7.25	21	No
5,471,316	4.92	36	Yes
8,729,043	4.5	50	No

using exponential attenuation functions derived from the Lambert–Beer law and accounting for incident angle, biomass concentration, and reactor depth. The general forms of these functions are provided in our previous work [26], while their implementation within ANSYS Fluent was achieved via a User-Defined Function (UDF), with the full code provided in the Appendix of this study for reproducibility.

It should be noted that bubble-induced scattering effects were not explicitly included in the present light attenuation model. This simplification was made because biomass absorption is generally the dominant mechanism controlling light attenuation in dense algal cultures, whereas modelling bubble scattering would require more complex radiative transfer approaches and significantly increase computational cost. Nevertheless, bubble scattering and gas holdup are known to influence light availability, as highlighted in previous CFD–PBR studies [62,63], and we identify their explicit inclusion as an important avenue for future work.

The biomass growth rate r_X is calculated as:

$$r_X = \frac{J_{O_2} C_X M_X}{\upsilon_{O_2-X}} \tag{2}$$

where J_{O_2} is the oxygen production rate, C_X is the biomass concentration, M_X is the molar mass of biomass, and υ_{O_2-X} is the stoichiometric coefficient linking oxygen production to biomass growth. J_{O_2} is determined as:

$$J_{O_2} = \rho_M \frac{K}{K+G} \overline{\phi'}_{O_2} E_a G \mathscr{H}(G - G_C) \tag{3}$$

where ρ_M is the maximum energy yield for photon conversion, K is the half-saturation constant for light intensity, G is the local irradiance, and G_C is the compensation irradiance value, and $\mathscr{H}(G - G_C)$ is the Heaviside function that ensures $J_{O_2}$is nonzero only when $G > G_C$. The mass absorption coefficient (E_a) determines light absorption by the algal culture, while the photosynthetic efficiency factor ($\overline{\phi'}_{O_2}$) quantifies the conversion of absorbed light into oxygen production, linking light intensity to oxygen production. The parameter values used in Eqs. (2) and (3) are summarized in Table 2. These parameters were experimentally derived and validated in our previous study [26], which served as the baseline for the current CFD simulations.

The UDF was integrated into the Eulerian-Eulerian multiphase framework of ANSYS Fluent, allowing the coupling of hydrodynamic simulations with biological productivity. At each time step, the UDF updated the growth rate dynamically based on the local flow field and light distribution, providing a detailed prediction of biomass productivity across the reactor. The complete UDF code used for simulating biomass growth is provided in the Appendix for reference and reproducibility.

2.3. Simulation parameters and scenarios

The simulations investigated the hydrodynamic behaviour of a flat plate photobioreactor (PBR) under varying sparger configurations and air injection flow rates, analysing 20 cases that combined five sparger positions with four flow rates. The sparger positions, evenly distributed along the reactor width (Fig. 1), were designed to evaluate their impact on key hydrodynamic parameters, including velocity distribution, turbulence kinetic energy (TKE), mixing efficiency, and dead zone formation.

Air injection flow rates were selected to replicate experimental conditions, ensuring realistic interactions between sparger placement and airflow dynamics. Each configuration produced distinct mixing patterns, influencing bubble dispersion, turbulence intensity, and dead zone percentages—regions of stagnant flow that impede mixing efficiency. Dead zones were quantified across all 20 runs to identify suboptimal mixing areas and their dependence on sparger position and airflow rate. The analysis revealed the critical role of sparger placement in reducing dead zones and enhancing mixing uniformity.

The hydrodynamic performance, assessed through velocity fields, TKE distributions, and dead zone percentages, provided key insights into the interplay between sparger configuration and reactor efficiency. These findings form a robust basis for optimizing flat plate PBR design, offering strategies to enhance mixing and reduce stagnation, ultimately contributing to improved algal productivity in future studies.

Table 2
Parameter values used in the biomass productivity model (Eqs. (2) and (3)), adapted from our previous experimental study [26].

Symbol	Description	Value	Unit
ρ_M	Maximum energy yield for photon conversion	0.8	–
K	Half-saturation constant for light intensity	90	$\mu mol_{h\nu} \cdot s^{-1} \cdot m^{-2}$
G_c	Compensation irradiance value	1.5	$Mmol_{h\nu} \cdot s^{-1} \cdot m^{-2}$
E_a	Mass absorption coefficient	105	$m^2 \cdot kgX^{-1}$
$\overline{\phi'}_{O_2}$	Mole quantum yield for *Z*-scheme photosynthesis	1.1×10^{-7}	$molO_2 \cdot \mu mol_{h\nu}^{-1}$
υ_{O_2-X}	Stoichiometric coefficient linking O_2 production to biomass	1.44	$molO_2 \cdot molX^{-1}$
M_X	C-molar mass of biomass	0.024	$kg_X \cdot mol_X^{-1}$

2.4. Model validation

The CFD model was validated by comparing simulated *A. platensis* productivity with experimental data under identical conditions. Validation was performed for the middle sparger position at four distinct air injection flow rates, mirroring the experimental setup. Simulated productivity was calculated using a User-Defined Function (UDF) that dynamically coupled hydrodynamics with biomass growth kinetics, incorporating irradiance distribution and local flow characteristics.

The comparison demonstrated strong agreement between simulated and experimental productivity, confirming the model's accuracy in capturing the interactions between hydrodynamics and biological performance. This validation established the model as a reliable tool for further analysis of sparger configurations and operational optimization in flat plate photobioreactors.

3. Results and discussion

This section provides a comprehensive analysis of the Computational Fluid Dynamics (CFD) simulation results, validated against experimental data to establish their reliability. Key hydrodynamic metrics, including velocity distribution, turbulence kinetic energy (TKE), and mixing efficiency, are examined to elucidate the influence of sparger configurations and air injection flow rates on the reactor's performance. The findings highlight critical trends and relationships, offering insights into the mechanisms driving mixing and flow uniformity within the photobioreactor. By systematically evaluating these parameters, the study identifies optimal configurations for enhancing hydrodynamic behaviour, ensuring a balanced discussion of both numerical results and their implications for reactor design and operation.

3.1. Validation of simulation results

The accuracy and reliability of the Computational Fluid Dynamics (CFD) model were validated by comparing simulated biomass productivity with experimental data previously reported by our research group [26]. The validation was carried out for the center sparger position under four different aeration rates (0.17, 0.19, 0.21, and 0.23 vvm), replicating the experimental conditions precisely. Both experimental and simulated biomass productivity data exhibited similar trends, with productivity consistently increasing as the aeration rate rose (Fig. 2). This trend alignment demonstrates the capability of the CFD model to accurately represent key hydrodynamic-biological interactions observed

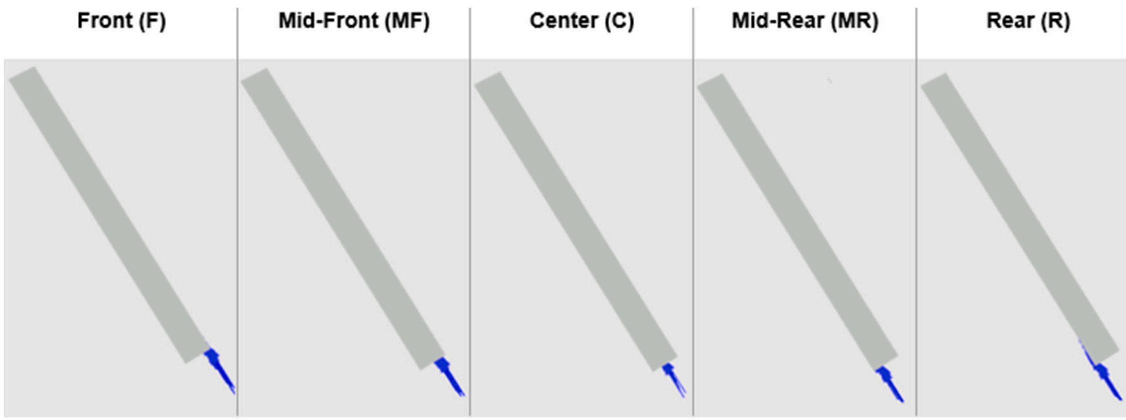

Fig. 1. Schematic representation of sparger positions in the photobioreactor, labelled as F (Front), MF (Mid-Front), C (Center), MR (Mid-Rear), and R (Rear). Blue arrows indicate the direction of air injection. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

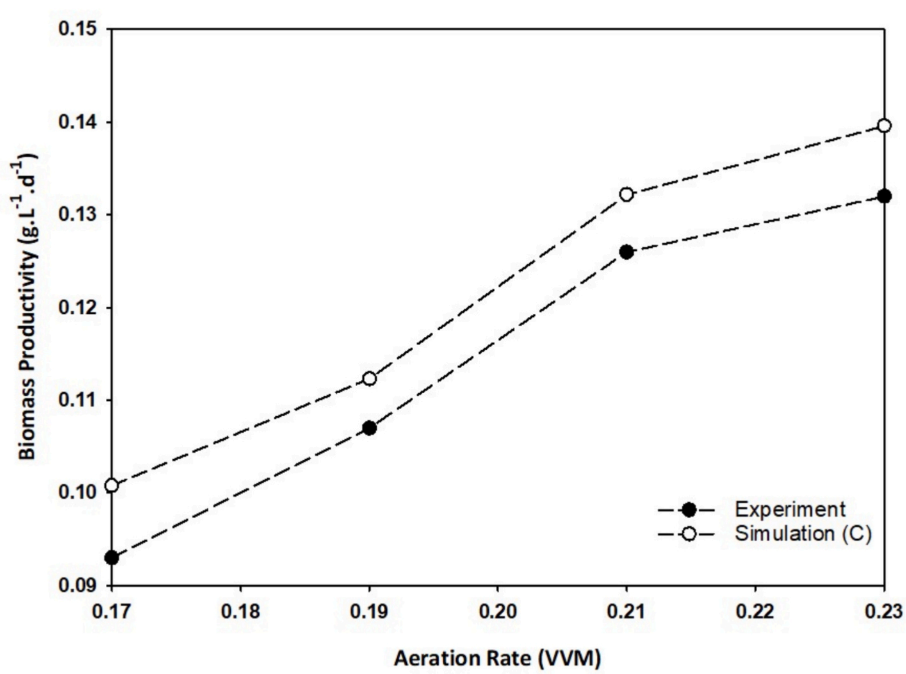

Fig. 2. Biomass productivity (g.L^{-1}.d^{-1}) comparison between experimental and simulation results under varying aeration rates (vvm).

experimentally.

The quantitative comparison indicated a maximum discrepancy of approximately 8.4 %, with CFD predictions slightly overestimating experimental productivity. This overestimation likely stems from the idealized assumptions inherent to the CFD simulation, such as perfectly uniform illumination, homogeneous nutrient distribution, and simplified hydrodynamic conditions, which might not fully capture the micro-scale complexities and localized shear stresses encountered in real-world experiments. Furthermore, the experimental observations indicated that aeration rates exceeding 0.23 vvm could induce mechanical stress detrimental to microalgal growth, a biological response not currently included in the simulation framework.

Statistical analysis further reinforced model reliability, yielding a high coefficient of determination ($R^2 = 0.81$) and a low mean relative error (MRE = 6.5 %). These metrics collectively signify strong agreement between simulation predictions and experimental outcomes, validating the effectiveness of the employed User-Defined Function (UDF) in dynamically coupling fluid flow patterns with microalgal growth kinetics. The validated CFD-UDF model thus emerges as a robust computational tool, providing detailed and reliable insights into hydrodynamic-biological interactions within photobioreactors. This validation lays a strong foundation for subsequent investigations aimed at optimizing sparger placement and enhancing photobioreactor performance in practical, scalable operations.

The experimental productivity data shown in Fig. 2 are presented as single representative runs without replicates, as resource and seasonal constraints precluded repeat experiments. Nevertheless, the close agreement between experimental and CFD-predicted productivity (mean relative error 6.5 %) provides confidence in the robustness of the dataset. Furthermore, the observed aeration–productivity relationship aligns with previous flat panel PBR studies, where moderate aeration enhanced mixing and light distribution but excessive aeration induced physiological stress [26,64]. Comparable validation discrepancies (typically within 5–10 %) have also been reported in CFD–PBR studies [65–67], supporting the reliability of the present results despite the absence of replicates.

It should be noted that validation was limited to the center sparger position due to the availability of experimental data. While this configuration provides a representative benchmark, extending validation to additional sparger placements would further strengthen the generalizability of the CFD framework and represents an important avenue for future experimental work.

3.2. Analysis of velocity distribution and mixing patterns

The velocity distribution analysis, as illustrated in Fig. 3 and supported by the quantitative summary in Table 3, reveals the critical influence of sparger placement and aeration rate on hydrodynamic performance in the inclined flat plate photobioreactor. While increasing aeration rates generally enhance internal mixing, the extent and uniformity of velocity distribution vary substantially with sparger position, indicating that spatial flow development is strongly dependent on injection location.

To support the visual insights from the velocity contours, a quantitative analysis of key hydrodynamic parameters— domain-averaged turbulence kinetic energy (TKE), radial velocity, and dead zone fraction—is presented in Table 3. The data is organized to highlight the comparative performance of all five sparger configurations across four aeration rates, using conditional formatting to visually emphasize performance gradients. The dead zone fraction is defined as the proportion of reactor volume where fluid velocity falls below a critical threshold, representing areas of insufficient mixing and potential stagnation. For this study, a threshold velocity of 0.02 m·s^{-1} was employed, consistent with standards in similar hydrodynamic analyses of photobioreactors [68,69]. This threshold delineates the minimum velocity required to sustain adequate mixing, prevent sedimentation of microalgae, and maintain optimal culture suspension within the photobioreactor.

Among the tested configurations, the rear-most sparger position (R) consistently generates the most homogeneous and spatially extensive velocity fields, particularly at 0.21 and 0.23 vvm. This configuration benefits from the reactor's inclined geometry, where rising air bubbles

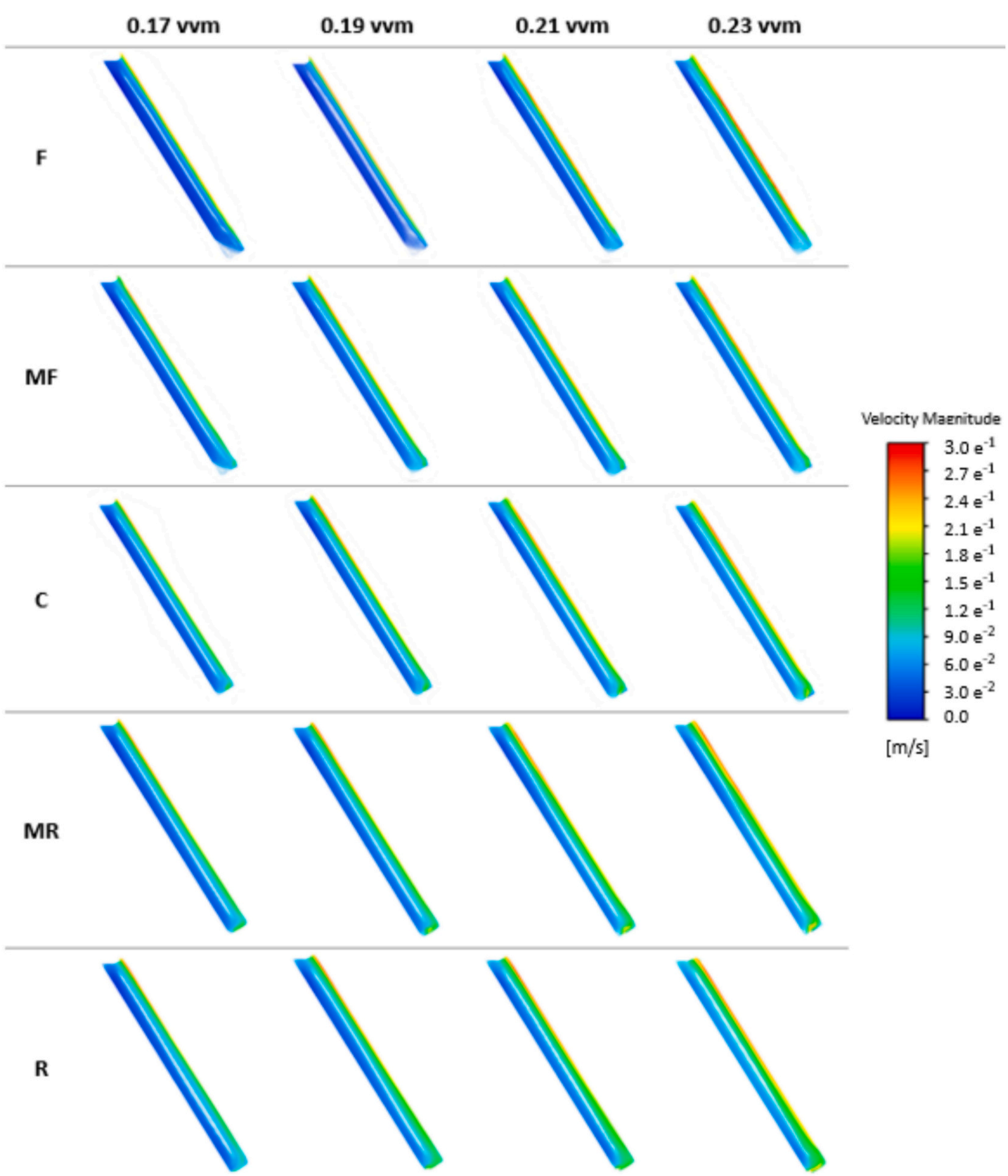

Fig. 3. Velocity magnitude contours for sparger positions F (Front), MF (Mid-Front), C (Center), MR (Mid-Rear), and R (Rear) at air injection rates of 0.17–0.23 vvm. Velocity magnitudes are represented by the colour scale on the right (m·s^{-1}).

Table 3
Quantitative analysis of turbulent kinetic energy (TKE), radial velocity, and dead zone fraction for various sparger positions (Rear, Mid-Rear, Center, Mid-Front, Front) at different aeration rates (vvm).

Sparger position	Aeration rate (vvm)	TKE × 10^{-3} ($m^2 \cdot s^{-2}$)	Radial velocity ($m \cdot s^{-1}$)	Dead zone fraction (%)
Rear (R)	0.23	4.51	0.134	15.56
	0.21	4.32	0.125	18.23
	0.19	4.17	0.112	20.54
	0.17	3.81	0.102	28.62
Mid-Rear (MR)	0.23	4.46	0.131	18.26
	0.21	4.29	0.119	20.72
	0.19	4.07	0.110	21.88
	0.17	3.77	0.100	28.90
Center (C)	0.23	4.39	0.126	22.39
	0.21	4.24	0.111	24.05
	0.19	4.01	0.104	25.35
	0.17	3.73	0.092	29.85
Mid-Front (MF)	0.23	4.26	0.112	25.30
	0.21	4.13	0.105	26.28
	0.19	3.93	0.090	27.93
	0.17	3.65	0.075	31.16
Front (F)	0.23	4.18	0.106	28.73
	0.21	4.02	0.103	29.01
	0.19	3.84	0.087	30.98
	0.17	3.53	0.068	32.29

are assisted by gravitational directionality, producing extended circulation loops that span the full reactor height. Notably, the rear position at 0.21 vvm achieves an average radial velocity of 0.125 $m \cdot s^{-1}$, nearly identical to the center position (C) at 0.23 vvm (0.126 $m \cdot s^{-1}$), but with a lower dead zone fraction (18.23 % vs. 22.39 %), suggesting improved flow efficiency at reduced energy input.

In contrast, the center sparger configuration delivers concentrated vertical momentum near the injection path, resulting in limited lateral spread and less engagement of peripheral zones. This is evident in the velocity contours, which display strong upward jets but diminished velocity gradients near reactor boundaries. Although effective at higher aeration rates, the center position underperforms compared to the rear when evaluated on energy-to-performance ratio.

Sparger placements toward the front face of the reactor (MF and F) exhibit suboptimal flow regimes. These configurations are characterized by narrow flow columns and significant low-velocity zones, as visualized by persistent blue regions in the contours. Even at 0.23 vvm, position F yields a radial velocity of only 0.106 $m \cdot s^{-1}$ and a dead zone fraction of 28.73 %, indicating inefficient mixing and poor fluid turnover. This likely results from bubbles being injected against the reactor's inclination, which reduces their vertical reach and inhibits full-cavity circulation. Consequently, stagnant regions become prevalent, which may negatively affect nutrient distribution, light exposure, and ultimately biomass productivity [70].

These findings are consistent with prior CFD–PBR sparger studies, which have similarly highlighted the sensitivity of flow uniformity and dead zone formation to sparger placement and geometry [31,34,62]. For example, Ali et al. [38] demonstrated that sparger design parameters (hole size, number, and gas flow rate) critically affect hydrodynamics and light transmission in flat-plate PBRs, with suboptimal configurations leading to stagnant regions and inefficient circulation. Likewise, Kulkarni et al. [45] showed that sparger type and distribution determine pressure profiles and bubble release uniformity, thereby influencing reactor mixing efficiency. While most previous studies investigated vertical flat-plate systems, the present work reveals that in inclined PBRs the effect of sparger placement is even more pronounced due to gravity-assisted bubble rise, underscoring the importance of optimized rearward configurations for minimizing dead zones and enhancing overall mixing.

Collectively, the results in this section demonstrate that sparger position plays a pivotal role in shaping internal flow structures and mixing uniformity in inclined flat plate photobioreactors. Rearward configurations generally promote more distributed and higher-magnitude flow fields, while front-oriented positions are associated with increased stagnation and reduced circulation coverage. However, while preliminary comparisons suggest potential advantages in terms of velocity distribution and dead zone minimization, a more comprehensive evaluation—particularly incorporating turbulence intensity and overall hydrodynamic efficiency—is necessary before identifying the optimal sparger configuration. The following sections address these aspects in greater detail to support a robust optimization framework.

3.3. Evaluation of turbulence kinetic energy (TKE)

Turbulence kinetic energy (TKE) serves as a critical metric for characterizing mixing intensity and energy dissipation within the photobioreactor, directly influencing microalgal suspension, nutrient transport, and overall biomass productivity [71,72]. Fig. 4 illustrates the spatial distribution of TKE across all five sparger positions (F to R) and four aeration rates, providing a visual reference for the zones of high and low turbulence intensity within the inclined photobioreactor.

The TKE contours reveal that turbulence is generally concentrated in regions immediately surrounding the sparger outlet, with intensity and

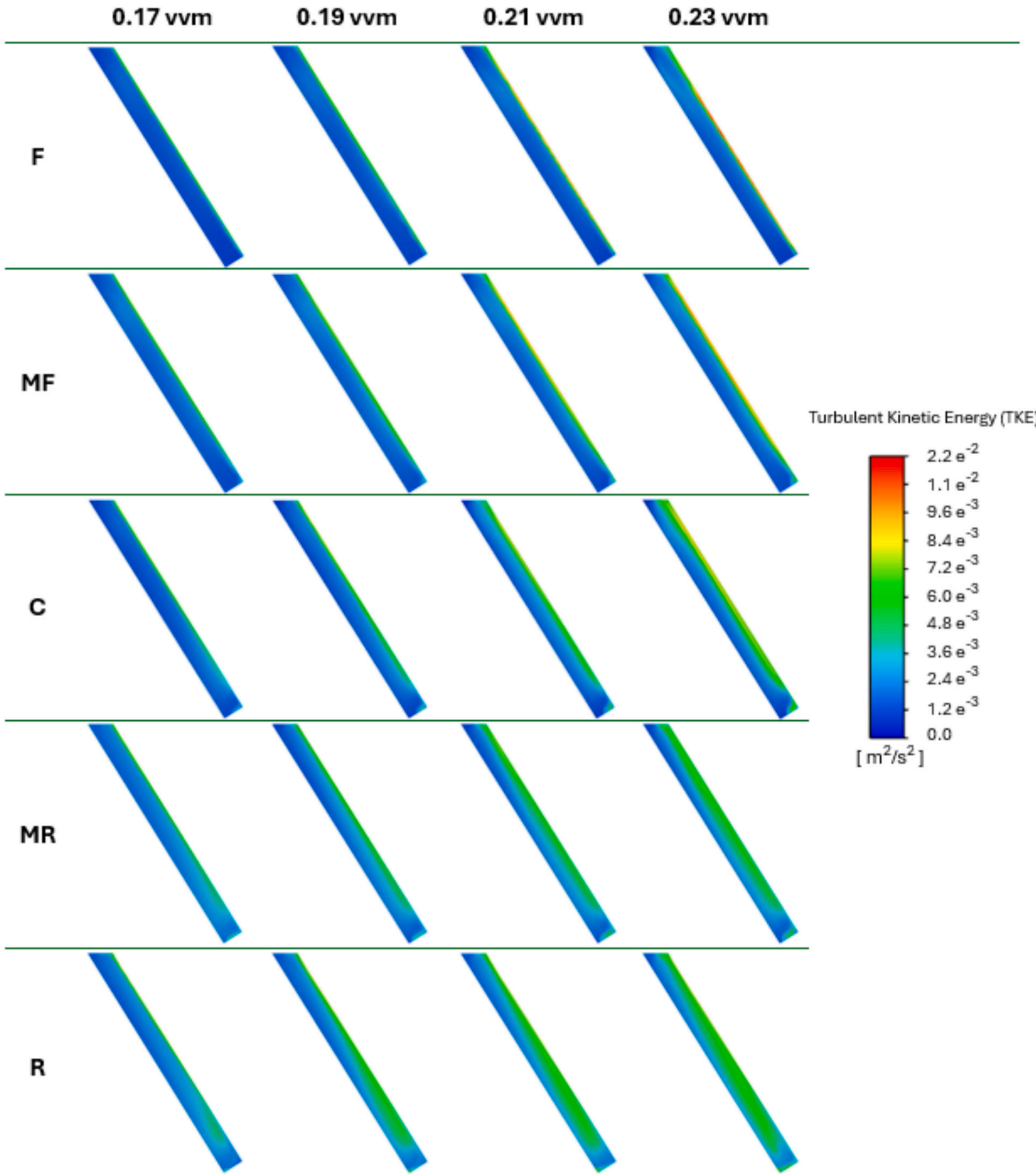

Fig. 4. Turbulent kinetic energy (TKE) distribution in PBRs for sparger positions F (Front), MF (Mid-Front), C (Center), MR (Mid-Rear), and R (Rear) at air injection rates of 0.17–0.23 vvm. TKE values are represented by the colour scale on the right ($m^2 \cdot s^{-2}$). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

spread strongly dependent on sparger location. At higher aeration rates (0.23 vvm), R and MR positions exhibit broader and more vertically extensive turbulence zones, particularly along the rear plane of the reactor. This spatial development aligns with the natural flow path induced by the reactor's inclination, allowing air bubbles to travel longer paths and generate sustained shear throughout the vertical axis.

In contrast, positions C, MF, and F produce more localized and narrower TKE regions, especially at lower aeration rates. These sparger placements tend to confine turbulence near the point of air injection, with less propagation into peripheral zones, resulting in a more constrained and inefficient mixing field. At 0.17 and 0.19 vvm, the blue-to-green regions in the F and MF configurations remain limited to the lower front corner of the reactor, leaving large volumes unaffected by active turbulence. This stagnant behaviour can lead to sedimentation and nutrient stratification, particularly problematic in high-density microalgal cultures [73].

The visual interpretation is further substantiated by Table 3, where average TKE values confirm the trends observed in the contours. At 0.23 vvm, the rear position (R) yields the highest average TKE (4.51×10^{-3} $m^2 \cdot s^{-2}$), followed closely by mid-rear (MR: 4.46×10^{-3}), with values declining progressively toward the front (F: 4.18×10^{-3}). Even at lower aeration rates, such as 0.21 vvm, position R maintains elevated turbulence (4.32×10^{-3}), comparable to center position C (4.39×10^{-3}) at 0.23 vvm rate. This suggests that rearward sparger placement can maintain turbulent mixing even under reduced airflow conditions, offering potential operational benefits in energy efficiency.

Importantly, previous experimental work using the center sparger position showed that exceeding 0.23 vvm led to significant physiological stress in *A. platensis*, notably reducing effective quantum yield due to increased shear exposure [26]. These findings underscore the need to balance turbulence generation with biological tolerance. The ability of position R to achieve comparable TKE at lower aeration rates supports its suitability as an energy-efficient and biologically safer alternative.

From a spatial standpoint, the rear configuration distributes turbulence more uniformly throughout the central and upper reactor regions, while front configurations exhibit restricted turbulence that does not effectively reach upper or rear zones. This distributional disparity is critical, as uniform turbulence contributes to homogenous light exposure, efficient gas-liquid transfer, and reduced biomass settlement, all vital for maximizing photobioreactor productivity [27,74].

The magnitude of TKE observed in this study ($3.5–4.5 \times 10^{-3}$ $m^2 \cdot s^{-2}$) falls within the lower end of values typically reported for flat-panel photobioreactors operating under moderate aeration [72,75,76]. Previous studies have indicated that excessive turbulence can induce mechanical stress and reduce photosynthetic efficiency in shear-sensitive strains [77,78], highlighting the importance of balancing mixing intensity with biological tolerance. Thus, while the rear sparger configuration promoted higher TKE compared to other positions, the absolute values remained within a biologically tolerable range for *A. platensis*. This supports the conclusion that sparger placement can be used to enhance mixing efficiency without exceeding shear thresholds detrimental to cell performance.

Overall, the TKE patterns shown in Fig. 4 reinforce the conclusions from the velocity distribution analysis in Section 3.2, highlighting how sparger location influences not just flow speed but also energy dissipation mechanics throughout the reactor volume. However, while rearward positions demonstrate promising turbulence profiles, further analysis in the following section (3.4) is necessary to fully integrate TKE with radial velocity and dead zone metrics for a comprehensive assessment of hydrodynamic performance.

3.4. Hydrodynamic and productivity assessment

Building upon the preceding velocity and turbulence analyses, a final quantitative assessment was conducted using dead zone fraction and simulated biomass productivity to holistically evaluate the hydrodynamic performance of each sparger configuration. These two parameters collectively reflect both fluid mixing efficiency and its biological impact on cultivation performance.

As presented in Table 3, the rear sparger position (R) consistently exhibited the lowest dead zone fractions across all aeration rates, suggesting more effective fluid circulation and reduced stagnant volume. Notably, at 0.21 vvm, position R achieved a dead zone fraction of 18.23 %, outperforming the center position (C) at 0.23 vvm (22.39 %), despite operating at a lower airflow rate. Combined with its comparable radial velocity (0.125 vs. 0.126 $m \cdot s^{-1}$) and slightly lower TKE, this suggests that position R can achieve mixing conditions similar to those of position C at a reduced aeration rate. To further contextualize these hydrodynamic advantages, Fig. 5 presents the predicted biomass productivity, simulated using a CFD-integrated growth model for *A. platensis* across all sparger positions and aeration rates.

The simulation results indicate that biomass productivity increases with aeration rate for all configurations, yet the benefits of sparger placement become more pronounced at higher flow rates. At 0.21 and 0.23 vvm, the rear and mid-rear configurations (R and MR) consistently outperformed others. In particular, position R at 0.21 vvm achieved closely matched productivity to position C at 0.23 vvm, demonstrating similar hydrodynamic performance with reduced aeration.

This outcome holds significant practical implications. Prior experimental studies, conducted at position C, identified 0.23 vvm as the upper tolerable limit for aeration, beyond which mechanical stress negatively affected *A. platensis* physiology, notably reducing effective quantum yield [26]. The ability of position R to replicate such conditions at 0.21 vvm—a safer and more energy-efficient rate—suggests a strategic advantage in sparger relocation for optimizing cultivation performance without compromising cellular health.

In summary, this final analysis demonstrates that position R at 0.21 vvm offers a compelling balance between hydrodynamic performance, energy efficiency, and predicted biological productivity. These results not only validate the potential of rear sparger configurations in inclined

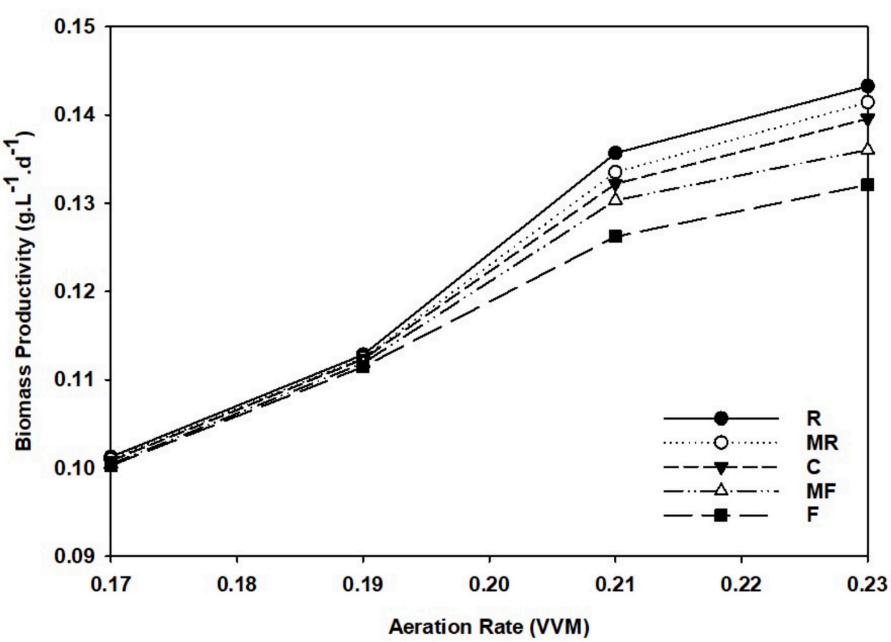

Fig. 5. Simulated biomass productivity for various sparger positions (Rear [R], Mid-Rear [MR], Center [C], Mid-Front [MF], Front [F]) at aeration rates ranging from 0.17 to 0.23 VVM.

PBRs but also provide strong computational guidance for future experimental optimization and scale-up.

4. Conclusion

This study developed and validated a CFD model, coupled with a user-defined function (UDF), to simulate hydrodynamic behaviour and microalgal growth in an inclined flat plate photobioreactor (PBR). The model demonstrated strong agreement with experimental data ($R^2 = 0.81$, MRE = 6.5 %), confirming its predictive reliability despite minor deviations due to simplified assumptions. Model validation was conducted using the central sparger position (C) across four aeration rates (0.17–0.23 vvm), corresponding directly to prior experimental configurations.

Key results revealed that repositioning the sparger to the rear (R) allows for nearly equivalent hydrodynamic performance at a reduced aeration rate. Specifically, at 0.21 vvm, position R achieved radial velocity and turbulence values comparable to position C at 0.23 vvm, while also reducing the dead zone fraction from 22.39 % to 18.23 %. This has clear operational significance, as previous experimental studies have shown that aeration beyond 0.23 vvm can negatively impact *A. platensis* through shear-related physiological stress.

Simulated biomass productivity, predicted through CFD-integrated growth kinetics, further reinforced the hydrodynamic analysis. Rearward sparger configurations (R and MR) demonstrated the highest productivity at elevated aeration rates, confirming their suitability under intensive cultivation conditions. At lower aeration rates, productivity differences were minimal, suggesting that strategic sparger placement becomes increasingly important under higher airflow demands.

Front-facing configurations (F and MF) consistently exhibited the weakest performance, characterized by higher dead zone fractions, lower radial velocities, and inefficient use of the reactor's inclined geometry. In contrast, the rear sparger configuration aligned most effectively with the reactor's natural flow dynamics, facilitating uniform circulation, enhanced turbulence, and energy-efficient mixing.

Looking ahead, future studies should aim to experimentally validate the computational predictions presented in this work, particularly with respect to shear-related effects on *A. platensis* under optimized flow conditions. In parallel, further CFD investigations could explore alternative sparger designs and configurations, in addition to baffle integration, to enhance mixing efficiency and minimize dead zones. Importantly, while this work focused on a laboratory-scale system, the validated CFD framework is readily extendable to larger PBR geometries. Scalability challenges, including sustaining uniform light distribution, minimizing shear stress, and maintaining mixing efficiency, remain key areas for investigation, and the present findings on sparger placement provide a foundation for addressing them. Finally, while this study qualitatively demonstrated energy-saving potential through reduced aeration requirements, future work should include quantitative estimates of energy savings to enable clearer techno-economic assessments.

In conclusion, the validated CFD-UDF framework developed here provides a powerful tool for optimizing photobioreactor configurations. By demonstrating that rear sparger placement at a reduced aeration rate can maintain high hydrodynamic and biological performance, this work offers actionable insights for improving energy efficiency, scalability, and cultivation outcomes in inclined flat plate photobioreactors.

Symbols

Greek letters

Symbol	Description	Unit
$\boldsymbol{\nu_{O_2-X}}$	stoichiometric coefficient of the oxygen production	$molO_2.mol_x^{-1}$
$\overline{\boldsymbol{\phi}}'_{O_2}$	photosynthetic efficiency factor for oxygen production	$molO_2.\mu mol_{h\nu}^{-1}$
$\boldsymbol{\rho_M}$	maximum energy yield for photon conversion	–

(continued on next column)

Latin letters

Symbol	Description	Unit
$\boldsymbol{C_X}$	Biomass concentration	$kg_X\cdot m^{-3}$
$\boldsymbol{E_a}$	mass absorption coefficient	$m^2\cdot kgX^{-1}$
$\boldsymbol{G}$	local irradiance in culture	$\mu mol_{h\nu}\cdot s^{-1}\cdot m^{-2}$
$\boldsymbol{G_d}$	diffuse component of local irradiance	$\mu mol_{h\nu}\cdot s^{-1}\cdot m^{-2}$
$\boldsymbol{G_D}$	direct component of local irradiance	$\mu mol_{h\nu}\cdot s^{-1}\cdot m^{-2}$
$\boldsymbol{J_{O_2}}$	Specific rate of O_2 production or consumption	$molO_2\cdot kg_X^{-1}\cdot s^{-1}$
$\boldsymbol{K}$	half saturation constant for photosynthesis	$\mu mol_{h\nu}\cdot s^{-1}\cdot m^{-2}$
$\boldsymbol{M_X}$	molar mass for the biomass	$kg_X\cdot mol_X^{-1}$
$\boldsymbol{r_x}$	biomass volumetric growth rate (productivity)	$kg_X\cdot m^{-3}\cdot s^{-1}$

CRediT authorship contribution statement

Behnam Amanna: Writing – original draft, Visualization, Software, Methodology, Formal analysis, Conceptualization. **Parisa A. Bahri:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Navid R. Moheimani:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no financial or personal relationships with any individuals or organizations that could be perceived as influencing the work presented in this article.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.algal.2025.104423.

Data availability

The UDF is provided in the Appendix. Validation data were published in Algal Research: https://doi.org/10.1016/j.algal.2025.104035.

References

[1] F. Ameen, S. AlNadhari, A.A. Al-Homaidan, Marine microorganisms as an untapped source of bioactive compounds, Saudi J. Bio. Sci. 28 (1) (2021) 224–231.

[2] R. Fernandes, J. Campos, M. Serra, J. Fidalgo, H. Almeida, A. Casas, D. Toubarro, A.I. Barros, Exploring the benefits of phycocyanin: from Spirulina cultivation to its widespread applications, Pharmaceuticals 16 (4) (2023) 592.

[3] J. Gumbo, C. Nesamvuni, A Review: Spirulina a Source of Bioactive Compounds and Nutrition, 2017.

[4] N. Seyidoglu, S. Inan, C. Aydin, A prominent superfood: Spirulina platensis, in: Superfood and Functional Food the Development of Superfoods and Their Roles as Medicine 22, 2017, pp. 1–27.

[5] D. Liestianty, I. Rodianawati, R.A. Arfah, A. Assa, Patimah, Sundari, Muliadi(s), Nutritional analysis of spirulina sp to promote as superfood candidate, in: IOP Conference Series: Materials Science and Engineering, IOP Publishing, 2019, p. 012031.

[6] X. Wang, G. Jin, K. Pan, B. Zhu, Y. Li, Effects of fluctuating temperature in open raceway ponds on the biomass accumulation and harvest efficiency of Spirulina in large-scale cultivation, Environ. Sci. Pollut. Res. 28 (2021) 20794–20802.

[7] S. Villaró-Cos, J.L.G. Sánchez, G. Acién, T. Lafarga, Research trends and current requirements and challenges in the industrial production of spirulina as a food source, Trends Food Sci. Technol. 143 (2024) 104280.

[8] M.B. Zrimec, E. Sforza, L. Pattaro, D. Carecci, E. Ficara, A. Idà, N. Ferrer-Ledo, S. Canziani, S. Mangini, B. Lazar, Advances in Spirulina cultivation: techniques, challenges, and applications, in: Insights Into Algae-Fundamentals, Culture Techniques and Biotechnological Uses of Microalgae and Cyanobacteria, 2024.

[9] I. Ahmad, N. Abdullah, I. Koji, A. Yuzir, S.E. Muhammad, Evolution of photobioreactors: a review based on microalgal perspective. IOP Conference Series, Mater. Sci. Eng. (2021). IOP Publishing: 012004.

[10] SN Chanquia, G Vernet, S Kara, Photobioreactors for cultivation and synthesis: Specifications, challenges, and perspectives, Eng. Life Sci. 22 (12) (2022 Dec) 712–724.
[11] S. Rezvani, I. Saadaoui, H. Al Jabri, N.R. Moheimani, Techno-economic modelling of high-value metabolites and secondary products from microalgae cultivated in closed photobioreactors with supplementary lighting, Algal Res. 65 (2022) 102733.
[12] M. Borowitzka, Commercial-scale production of microalgae for bioproducts, in: Blue Biotechnology: Production and Use of Marine Molecules 1, 2018, pp. 33–65.
[13] R. Meier, Biological cycles in the transformation of solar energy into useful fuels, Sol. Energy Res. 23 (1955) 179–183.
[14] J. Assunção, F.X. Malcata, Enclosed "non-conventional" photobioreactors for microalga production: a review, Algal Res. 52 (2020) 102107.
[15] E. Kouhgardi, S. Zendehboudi, O. Mohammadzadeh, A. Lohi, I. Chatzis, Current status and future prospects of biofuel production from brown algae in North America: progress and challenges, Renew. Sust. Energ. Rev. 172 (2023) 113012.
[16] P. Darvehei, P.A. Bahri, N.R. Moheimani, Model development for the growth of microalgae: a review, Renew. Sust. Energ. Rev. 97 (2018) 233–258.
[17] V. Gernigon, M.A. Chekroun, A. Cockx, P. Guiraud, J. Morchain, How mixing and light heterogeneity impact the overall growth rate in photobioreactors, Chem. Eng. Technol. 42 (8) (2019) 1663–1669.
[18] M.A. Borowitzka, Algal physiology and large-scale outdoor cultures of microalgae of the chapter, in: The Physiology of Microalgae, Springer, 2016.
[19] Q. Béchet, A. Shilton, B. Guieysse, Modeling the effects of light and temperature on algae growth: state of the art and critical assessment for productivity prediction during outdoor cultivation, Biotechnol. Adv. 31 (8) (2013) 1648–1663.
[20] J. Legrand, A. Artu, J. Pruvost, A review on photobioreactor design and modelling for microalgae production, React. Chem. Eng. 6 (7) (2021) 1134–1151.
[21] V. Belohlav, E. Uggetti, J. García, T. Jirout, L. Kratky, R. Díez-Montero, Assessment of hydrodynamics based on computational fluid dynamics to optimize the operation of hybrid tubular photobioreactors, J. Environ. Chem. Eng. 9 (5) (2021) 105768.
[22] V. Belohlav, T. Jirout, L. Kratky, Optimization of hydrodynamics by installation of static mixer in flat panel photobioreactor, Chem. Eng. Trans. 86 (2021) 139–144.
[23] E.G. Nwoba, D.A. Parlevliet, D.W. Laird, A. Vadiveloo, K. Alameh, N. R. Moheimani, Can solar control infrared blocking films be used to replace evaporative cooling for growth of Nannochloropsis sp. in plate photobioreactors? Algal Res. 39 (2019) 101441.
[24] E.G. Nwoba, D.A. Parlevliet, D.W. Laird, K. Alameh, N.R. Moheimani, Pilot-scale self-cooling microalgal closed photobioreactor for biomass production and electricity generation, Algal Res. 45 (2020) 101731.
[25] E.G. Nwoba, D.A. Parlevliet, D.W. Laird, K. Alameh, N.R. Moheimani, Outdoor phycocyanin production in a standalone thermally-insulated photobioreactor, Bioresour. Technol. 315 (2020) 123865.
[26] B. Amanna, P.A. Bahri, G. Zhang, N.R. Moheimani, Enhancing Arthrospira platensis productivity by optimizing mixing rates in a self-cooling flat plate photobioreactor, Algal Res. 88 (2025) 104035.
[27] A.P. Carvalho, L.A. Meireles, F.X. Malcata, Microalgal reactors: a review of enclosed system designs and performances, Biotechnol. Prog. 22 (6) (2006) 1490–1506.
[28] M.A. Borowitzka, A. Vonshak, Scaling up microalgal cultures to commercial scale, Eur. J. Phycol. 52 (4) (2017) 407–418.
[29] E. Arenas, M. Rodriguez Palacio, A. Juantorena, S. Fernando, P. Sebastian, Microalgae as a potential source for biodiesel production: techniques, methods, and other challenges, Int. J. Energy Res. 41 (6) (2017) 761–789.
[30] K. Schediwy, A. Trautmann, C. Steinweg, C. Posten, Microalgal kinetics—a guideline for photobioreactor design and process development, Eng. Life Sci. 19 (12) (2019) 830–843.
[31] A. Azarpour, S. Zendehboudi, O. Mohammadzadeh, A.R. Rajabzadeh, I. Chatzis, A review on microalgal biomass and biodiesel production through co-cultivation strategy, Energy Convers. Manag. 267 (2022) 115757.
[32] J.P. Bitog, I.-B. Lee, C.-G. Lee, K.-S. Kim, H.-S. Hwang, S.-W. Hong, I.-H. Seo, K.-S. Kwon, E. Mostafa, Application of computational fluid dynamics for modeling and designing photobioreactors for microalgae production: a review, Comput. Electron. Agric. 76 (2) (2011) 131–147.
[33] P. Ranganathan, A.K. Pandey, R. Sirohi, A. Tuan Hoang, S.H. Kim, Recent advances in computational fluid dynamics (CFD) modelling of photobioreactors: design and applications, Bioresour. Technol. 350 (2022) 126920.
[34] B. Amanna, P.A. Bahri, N.R. Moheimani, Application of computational fluid dynamics in optimizing microalgal photobioreactors, Algal Res. 83 (2024) 103718.
[35] P.F. Del Olmo, F. Acién, J.M. Fernández-Sevilla, Analysis of productivity in raceway photobioreactor using computational fluid dynamics particle tracking coupled to a dynamic photosynthesis model, Bioresour. Technol. 334 (2021) 125226.
[36] I. Perner, C. Posten, J. Broneske, CFD optimization of a plate photobioreactor used for cultivation of microalgae, Eng. Life Sci. 3 (7) (2003) 287–291.
[37] Z. Su, R. Kang, S. Shi, W. Cong, Z. Cai, Study on the destabilization mixing in the flat plate photobioreactor by means of CFD, Biomass Bioenergy 34 (12) (2010) 1879–1884.
[38] H. Ali, J. Solsvik, J.L. Wagner, D. Zhang, K. Hellgardt, C.W. Park, CFD and kinetic-based modeling to optimize the sparger design of a large-scale photobioreactor for scaling up of biofuel production, Biotechnol. Bioeng. 116 (9) (2019) 2200–2211.
[39] A. Berouaken, R. Rihani, F.S. Marra, Study of sparger design effects on the hydrodynamic and mass transfer characteristics of a D-shape hybrid airlift reactor, Chem. Eng. Res. Des. 191 (2023) 66–82.
[40] G. Li, H. Li, G. Wei, X. He, S. Xu, K. Chen, P. Ouyang, X. Ji, Hydrodynamics, mass transfer and cell growth characteristics in a novel microbubble stirred bioreactor employing sintered porous metal plate impeller as gas sparger, Chem. Eng. Sci. 192 (2018) 665–677.
[41] Y.A. Lim, I. Ilankoon, M.N. Chong, S.C. Foo, Improving microalgae growth and carbon capture through micro-size bubbles generation in flat-panel photobioreactors: impacts of different gas sparger designs on mixing performance, Renew. Sust. Energ. Rev. 171 (2023) 113001.
[42] I.-H. Seo, I.-B. Lee, S.-W. Hong, J.P. Bitog, K.-S. Kwon, C.-G. Lee, Z.H. Kim, Evaluation of a photobioreactor performance grafting microalgal growth model and particle tracking technique using CFD, Trans. ASABE 57 (1) (2014) 121–139.
[43] U.-H. Yeo, I.-B. Lee, I.-H. Seo, R.-W. Kim, Identification of the key structural parameters for the design of a large-scale PBR, Biosyst. Eng. 171 (2018) 165–178.
[44] G.S. Bari, T.N. Suess, G.A. Anderson, S.P. Gent, Hydrodynamic and heat transfer effects of varying sparger spacing within a column photobioreactor using computational fluid dynamics, J. Fuel Cell Sci. Technol. 12 (1) (2015).
[45] A. Kulkarni, S.S. Roy, J. Joshi, Pressure and flow distribution in pipe and ring spargers: experimental measurements and CFD simulation, Chem. Eng. J. 133 (1–3) (2007) 173–186.
[46] M. Raeisossadati, N.R. Moheimani, D. Parlevliet, Red and blue luminescent solar concentrators for increasing Arthrospira platensis biomass and phycocyanin productivity in outdoor raceway ponds, Bioresour. Technol. 291 (2019) 121801.
[47] E.G. Nwoba, D.A. Parlevliet, D.W. Laird, K. Alameh, N.R. Moheimani, Sustainable phycocyanin production from Arthrospira platensis using solar-control thin film coated photobioreactor, Biochem. Eng. J. 141 (2019) 232–238.
[48] C. Herold, T. Ishika, E.G. Nwoba, S. Tait, A. Ward, N.R. Moheimani, Biomass production of marine microalga Tetraselmis suecica using biogas and wastewater as nutrients, Biomass Bioenergy 145 (2021) 105945.
[49] S. Banerjee, S. Dasgupta, D. Das, A. Atta, Influence of photobioreactor configuration on microalgal biomass production, Bioprocess Biosyst. Eng. 43 (8) (2020) 1487–1497.
[50] X. Gao, Advanced CFD Model of Multiphase Photobioreactors for Microalgal Derived Biomass Production, Doctoral Dissertation, Iowa State University, 2016.
[51] M. Terashima, M. So, R. Goel, H. Yasui, Determination of diffuser bubble size in computational fluid dynamics models to predict oxygen transfer in spiral roll aeration tanks, J Water Process Eng 12 (2016) 120–126.
[52] P. Ham, S. Bun, P. Painmanakul, K. Wongwailikhit, Effective analysis of different gas diffusers on bubble hydrodynamics in bubble column and airlift reactors towards mass transfer enhancement, Processes 9 (10) (2021) 1765.
[53] Y. Fayolle, A. Cockx, S. Gillot, M. Roustan, A. Héduit, Oxygen transfer prediction in aeration tanks using CFD, Chem. Eng. Sci. 62 (24) (2007) 7163–7171.
[54] B. Wu, Advances in the use of CFD to characterize, design and optimize bioenergy systems, Comput. Electron. Agric. 93 (2013) 195–208.
[55] F.R. Menter, Turbulence Modeling for Engineering Flows, Ansys, Inc, 2011.
[56] B.A. Cho, G.M. Teke, G.K. Gakingo, R.W.M. Pott, D. Zhang, CFD predictive simulations of miniature bioreactor mixing dynamics coupled with photo-bioreaction kinetics in transitional flow regime, Biochem. Eng. J. 214 (2025) 109585.
[57] G. Luzi, C. McHardy, Modeling and simulation of photobioreactors with computational fluid dynamics—a comprehensive review, Energies 15 (11) (2022) 3966.
[58] Y. Chisti, Large-scale production of algal biomass: raceway ponds, in: *Algae biotechnology: Products and processes*, Springer International Publishing, Cham, 2016, pp. 21–40.
[59] C.L. Hinterholz, D.E.G. Trigueros, A.N. Modenes, C.E. Borba, F.B. Scheufele, A. R. Schuelter, A.D. Kroumov, Computational fluid dynamics applied for the improvement of a flat-plate photobioreactor towards high-density microalgae cultures, Biochem. Eng. J. 151 (2019) 107257.
[60] X. Lu, J.Z. Tan, M.M. Maroto-Valer, Investigation of CO2 photoreduction in an annular fluidized bed photoreactor by MP-PIC simulation, Ind. Eng. Chem. Res. 61 (8) (2022) 3123–3136.
[61] J. Pruvost, J.-F. Cornet, Knowledge models for the engineering and optimization of photobioreactors, in: Microalgal Biotechnology: Potential and Production, 2012, pp. 181–224.
[62] B. Anye Cho, M.Á. de Carvalho Servia, E.A. del Río Chanona, R. Smith, D. Zhang, Synergising biomass growth kinetics and transport mechanisms to simulate light/dark cycle effects on photo-production systems, Biotechnol. Bioeng. 118 (5) (2021) 1932–1942.
[63] B. Anye Cho, Surrogate and Multiscale Modelling for (Bio) Reactor Scale-up and Visualisation, 2023.
[64] H. Qiang, A. Richmond, Productivity and photosynthetic efficiency of Spirulina platensis as affected by light intensity, algal density and rate of mixing in a flat plate photobioreactor, J. Appl. Phycol. 8 (1996) 139–145.
[65] S.J. Heydari, A. Jafari, S.M. Mousavi, M.A. Khanghah, Numerical study of microalgae cultivation process in a batch liquid foam-bed photobioreactor using CFD approach, Renew. Energy 223 (2024) 120054.
[66] R. Zhao, L. Wang, Q. Wang, S. Dai, Z. Han, X. Lei, X.-Z. Mao, Integrated model for design, optimization, scale-up, culture scheme, and production evaluation of a bioreactor and its application for improving algal biomass and carbon fixation in bioreactors, ACS Sustain. Chem. Eng. 10 (48) (2022) 15939–15949.
[67] J.C.M. Pires, Mathematical modeling and simulation applied to microalgae, in: I. A. Severo, et al. (Eds.), Microalgae Horizons: Fundamentals, Innovations, and Industrial Applications, Springer Nature Switzerland, Cham, 2025, pp. 87–104.
[68] L. Wang, Q. Wang, R. Zhao, Y. Tao, K.-Z. Ying, X.-Z. Mao, Novel flat-plate photobioreactor with inclined baffles and internal structure optimization to

improve light regime performance, ACS Sustain. Chem. Eng. 9 (4) (2021) 1550–1558.
[69] B.A. Guler, I. Deniz, Z. Demirel, S.S. Oncel, E. Imamoglu, Computational fluid dynamics modelling of stirred tank photobioreactor for Haematococcus pluvialis production: hydrodynamics and mixing conditions, Algal Res. 47 (2020) 101854.
[70] Y. Liu, Z. Liu, Z. Xiong, Y. Geng, D. Cui, S.G. Pavlostathis, H. Chen, Q. Luo, G. Qiu, Q. Dong, Synergistic optimization of baffles and aeration to improve the light/dark cycle of microalgae photobioreactor for enhanced nitrogen removal performance: computational fluid dynamics and experimental verification, Bioresour. Technol. 410 (2024) 131293.
[71] L. Xu, R. Liu, F. Wang, C.-Z. Liu, Development of a draft-tube airlift bioreactor for Botryococcus braunii with an optimized inner structure using computational fluid dynamics, Bioresour. Technol. 119 (2012) 300–305.
[72] J. Huang, F. Feng, M. Wan, J. Ying, Y. Li, X. Qu, R. Pan, G. Shen, W. Li, Improving performance of flat-plate photobioreactors by installation of novel internal mixers optimized with computational fluid dynamics, Bioresour. Technol. 182 (2015) 151–159.
[73] L. Zhao, Z. Tang, Y. Gu, Y. Shan, T. Tang, Investigate the cross-flow flat-plate photobioreactor for high-density culture of microalgae, Asia Pac. J. Chem. Eng. 13 (5) (2018) e2247.
[74] Y. Sun, D. Hu, H. Chang, S. Li, S.-H. Ho, Recent progress on converting CO2 into microalgal biomass using suspended photobioreactors, Bioresour. Technol. 363 (2022) 127991.
[75] L.-L. Wang, Y. Tao, X.-Z. Mao, A novel flat plate algal bioreactor with horizontal baffles: structural optimization and cultivation performance, Bioresour. Technol. 164 (2014) 20–27.
[76] J. Xu, J. Cheng, X. Lai, X. Zhang, W. Yang, J.-Y. Park, H. Kim, L. Xu, Enhancing microalgal biomass productivity with an optimized flow field generated by double paddlewheels in a flat plate photoreactor with CO2 aeration based on numerical simulation, Bioresour. Technol. 314 (2020) 123762.
[77] A. Gupta, Transport and Hydrodynamic Stresses in Turbulent Flows with Application to Photo-bioreactors, 2019.
[78] M.J.G.D.V. Barbosa, Microalgal Photobioreactors: Scale-up and Optimisation, Wageningen University and Research, 2003.